

LAB 06

PH142 Summer 2026

Announcements

- 7/27: Quiz15, Lab06
- 7/28: Quiz16, Lab07
- 7/29: Quiz17
- 7/30: Midterm 2
- 7/31: Quiz18, Data Project Part 2*

*Your group must meet with your assigned GSI before the due date



Lab06 Lecture Review

Distributions

- *Distributions* are used to model patterns in data
- We will cover three distributions:
 - Normal
 - Binomial
 - Poisson
- Based on what we know about the data's underlying distribution, we can calculate the mean, variance/SD, and probabilities

Lab06 Lecture Review

The Normal Distribution

The Normal distribution is a bell-shaped, symmetric distribution, with notation $\mathbf{X} \sim \mathbf{N}(\mu, \sigma)$

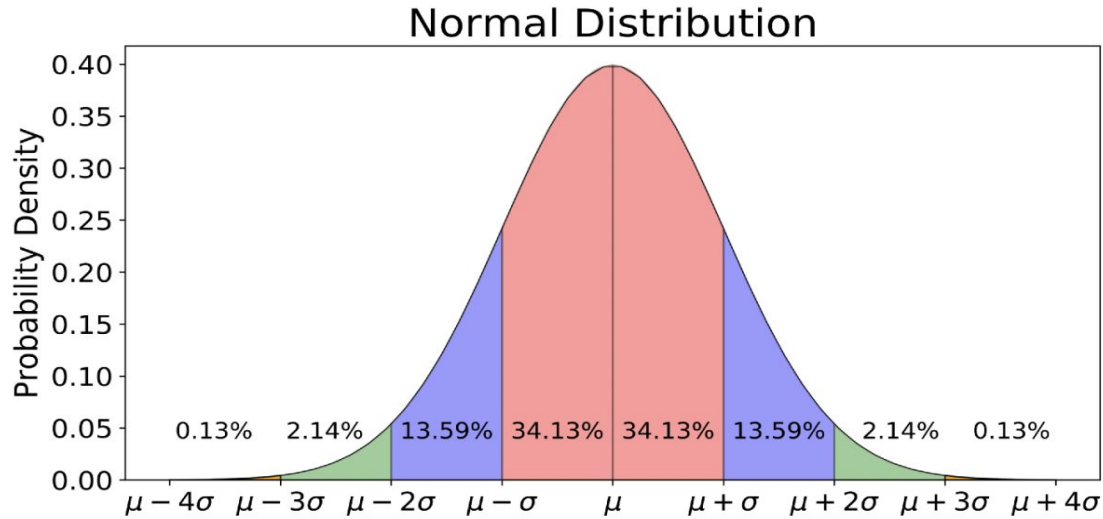
- μ = mean
- σ = standard deviation

R Functions:

- `pnorm()` → outputs the probability of value x or below
- `rnorm()` → generates random draws from the distribution
- `qnorm()` → outputs a quantile

Lab06 Lecture Review

The Normal Distribution



Lab06 Lecture Review

The Binomial Distribution

The Binomial distribution models discrete random variables (only whole numbers), with notation $\mathbf{X \sim Binom}(n, p)$

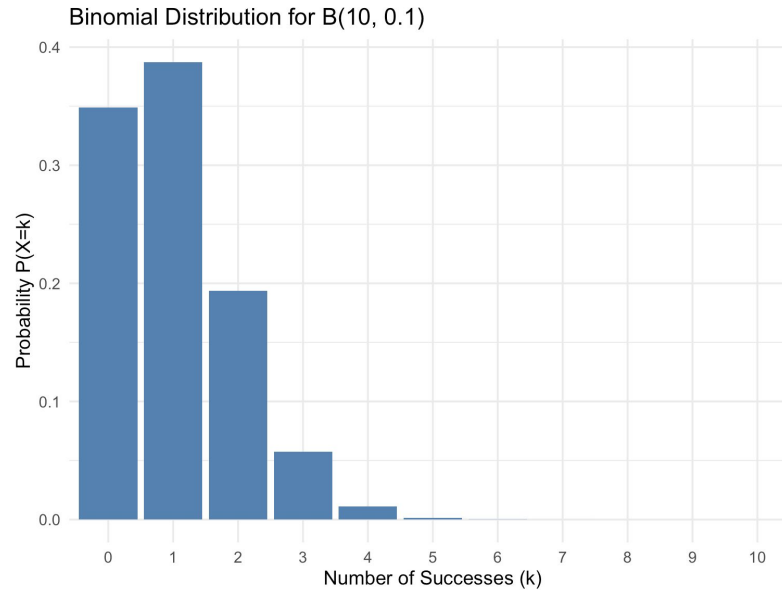
- n = total number of “trials”
- p = probability of success

R Functions:

- `pbinom()` → outputs the probability of value x or below
- `dbinom()` → outputs the probability of x successes exactly
- `rbinom()` → generates random draws from the distribution

Lab06 Lecture Review

The Binomial Distribution



Lab06 Lecture Review

The Poisson Distribution

A Poisson distribution describes the number of event occurrences in fixed, finite intervals of time or space, with notation **$X \sim \text{Pois}(\lambda)$**

- λ = the average number of events in an interval

R Functions:

- `ppois()` → outputs the probability of value x or below
- `dpois()` → outputs the probability of x events exactly
- `rpois()` → generates random draws from the distribution



LAB 06 Walkthrough

Lab Submission

- Follow the directions on the LAB06 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**